

TREND FOLLOWING TRADING¹

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Abstract

Trend following strategies have the reputation of being the holy grail of investment. This paper investigates the trend following strategy both from a theoretical and empirical perspective. This article also discusses conditional expected return and the drawdown with Value at Risk (VaR). VaR will not tell you how likely you are to experience changes in expected return for the securities in your portfolio. Such a number is very easy to estimate historically but not so easy to forecast i.e. how do you detect the next Bernard Madoff?

Keywords: finance, trend following, risk management

In the heart of modern portfolio theory (Markowitz, 1959 and Sharpe, 1964) lays the concept of trend following (Covel, 2004). Portfolio theory has more components than just trend following but trend following represent one of its cornerstones. The reason trend following is important is because trend following is concerned with expected returns also known as the arithmetic mean return. We can see in Exhibit-1 that the expected return for a long position in an uptrend (downtrend) is positive (negative). Any investor would prefer a positive expected return before a negative expected return. A trend also known as drift has two components high positive or negative expected return and low return volatility as seen in appendix-1. Both these conditions need to be fulfilled otherwise we will not have a price trend. We can model such trending behavior quite realistic by using a random walk model where price in the next period is a function of the price in the previous period plus a drift term plus a random return. However, Taleb (2007) explains that the problem with such a simplistic model is the unrealistic assumption of constant mean and volatility. The author calls such constrained and artificial casino behavior mild randomness.

¹A revised version of Davidsson, M. (2011) Trend following and expected return, *Economics, Management, and Financial Markets*, 6(2), 39–56.

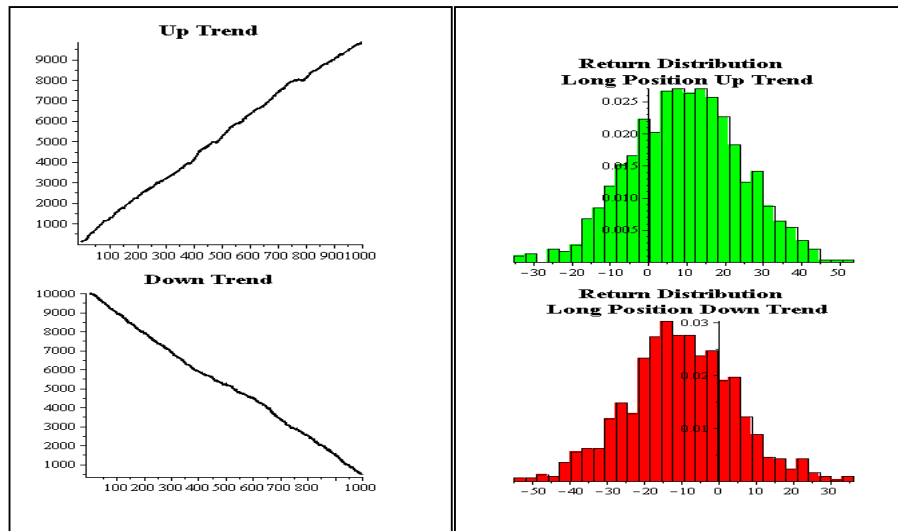


Exhibit 1. Trends and Expected Returns

Such point is highly important to make because most risk comes from changes in expected return. Traditional diversification need to be approached with a bit of caution since bond returns in general tend to be the only return that will not become negative during a market crash as seen in appendix-2. Bonds provide a good source of diversification even though the return on “risk free” government bonds has steadily been declining for the last 40 years.

There are basically three ways that changes in conditional expected return can be managed: i) Balance long and short positions (CAPM) ii) portfolio rebalancing iii) Trailing stop losses. Changes in conditional expected return do also play an important role when it comes to minimizing VaR. The idea is that changes in expected return are usually over looked as a source of portfolio risk i.e. the main focus is usually on return noise cancelation though portfolio diversification. Any diversified positive expected return portfolio can in an instance become a negative expected return portfolio i.e. positive correlation expected return. Again one way for an investor to make sure that the expected return remains positive is to constantly rebalance it i.e. replace all positions with negative expected return to positions with positive expected returns. Such constant rebalancing is not a walk in the park because it will induce transactions cost. Systematic risk i.e. market risk will also be minimized by applying a frequent portfolio rebalancing strategy.

Cont (2005) further points out another weakness with the simple random walk model called volatility clustering. Volatility in financial markets is not constant as we previous discussed but it is also not completely stochastic as assumed in a stochastic volatility model. The truth lays somewhere in between. Volatility tends to have stochastic jumps but there also exist a significant amount

of positive serial correlation in the volatility. The GARCH model has been created to model such behavior in empirical asset returns (Engle, 1982). Large changes tend to be followed by large changes and small changes tend to be followed by small changes (Mandelbrot, 1963). Serial correlation in returns often tends to be insignificant. Positive serial correlation in expected return and volatility is a contributing factor why we frequently see price trends in financial markets. Another important question that a trend following investor need to answer is; when does a trend stop being a trend i.e. when does the positive expected return become negative. I have illustrated such a point in appendix-3. At such a point a trend following investor need to exit his or her position. Previous studies such as Davidsson (2010) has shown that a trailing stop loss represent a natural tool for a trend following investor because as long as the trend is intact the trailing stop loss is not triggered. The benefit with using a trailing stop loss is that such a tool will help an investor to detect changes in expected return. A trailing stop loss will make sure that an investor remain disciplined i.e. it might be difficult to detect a change in expected return if you don't monitor your portfolio actively.

Most professional portfolio managers tend to argue that the same effect can be achieved by monitoring changes in the Sharpe or Treynor ratio of the portfolio. The benefit with such an approach is that the portfolio investor can further reduce return volatility around the trend (i.e. noise) by diversification. However, such return noise should have little effect on the probability that the trailing stop will be triggered given that they are set appropriately i.e. not too tight in order to capture changes in expected return. This has to do with the fact that by definition a trend has low return volatility otherwise it would not be a trend hence the return noise will be less of a problem. Depending on what investment strategy you use your optimal risk management strategy will look different. A trend following strategy (conditional risk adjusted expected return) will have lower return volatility hence a trailing stop loss would be more appropriate to capture changes in expected returns. A momentum strategy (expected return) will have larger return volatility hence a portfolio rebalancing strategy would be more suitable. If you would apply a momentum strategy with a trailing stop loss you run the risk of having noise get you out of your position. However, even though you most likely would be able to achieve a positive expected return. A pure portfolio optimization strategy is traditionally used in combination with portfolio rebalancing strategy. We can now load some empirical data in the form of monthly data for SP-1000 stocks (SP-400 MidCap and SP-600 SmallCap) for the period 2003 to 2010 in order to empirically investigate the trend following return premium. Please note that some of these stocks had missing observation so they were simply removed hence 884 stocks. The ten month conditional risk adjusted returns for these stocks can be found in appendix-4. We start by investigating a simple strategy where we take a long position in all stocks and where the exit point is done according to a trailing stop loss. The equity curve can be seen in Exhibit-2. We can see that the equity curve is smooth and upward sloping. In order to calculate percentage returns we need to note that it will cost an investor 22,182 to buy one contract of all the stocks at time 1. In the last period our portfolio is worth

approximately $22,182 + 10,000 = 32,182$ which means that the annual percentage increase is 6.4 percent given a position size equal to one contract and zero transaction cost. We assume that when the investor has exited one of his positions he is allowed to reenter such a position if we wish. We can see in the below Exhibit that only a small fraction of the trades are held until the last period.

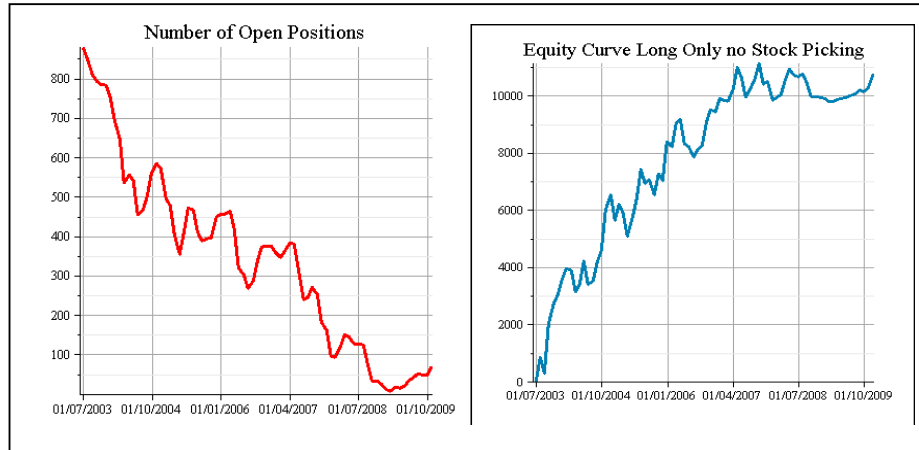


Exhibit 2. Trends and Expected Returns

We can now analyze what happens when we change some of the parameter values. In Exhibit-3 we can see how the profit/loss changes when we make changes in the number of stocks, transaction cost, trailing distance and initial capital. We can see that the higher the transaction cost is the lower our profit will become, which is natural. We can also see that the trail distance seem to be quadratic related to profit/loss. A too tight or too generous stop distance has a negative effect on profits. We can also see that the more initial capital we have the higher our profits will become given reasonable transaction costs.

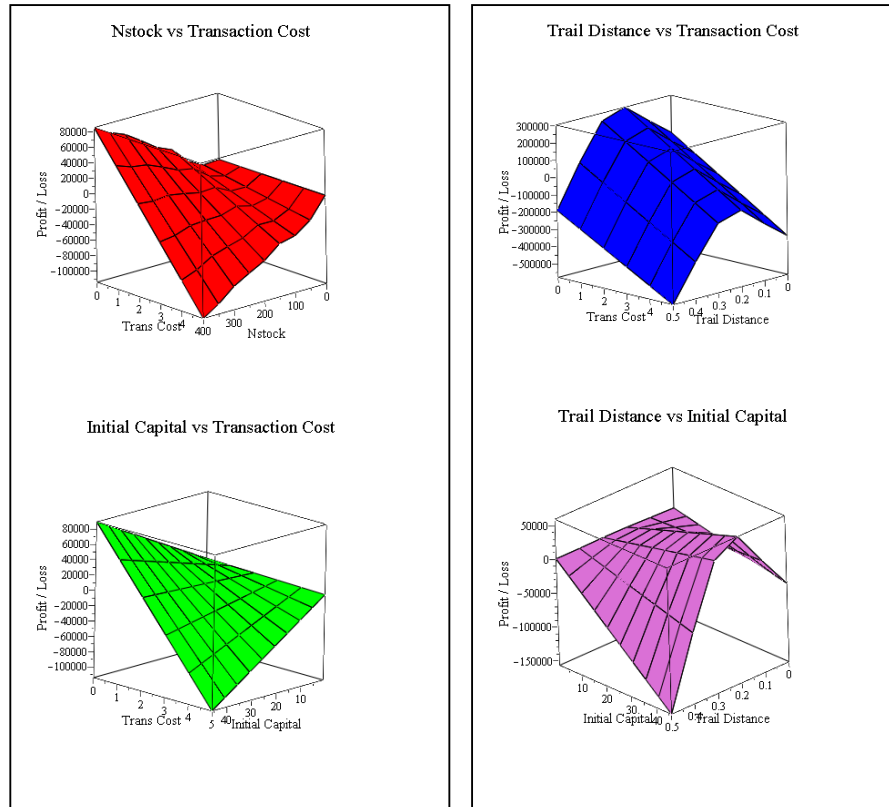
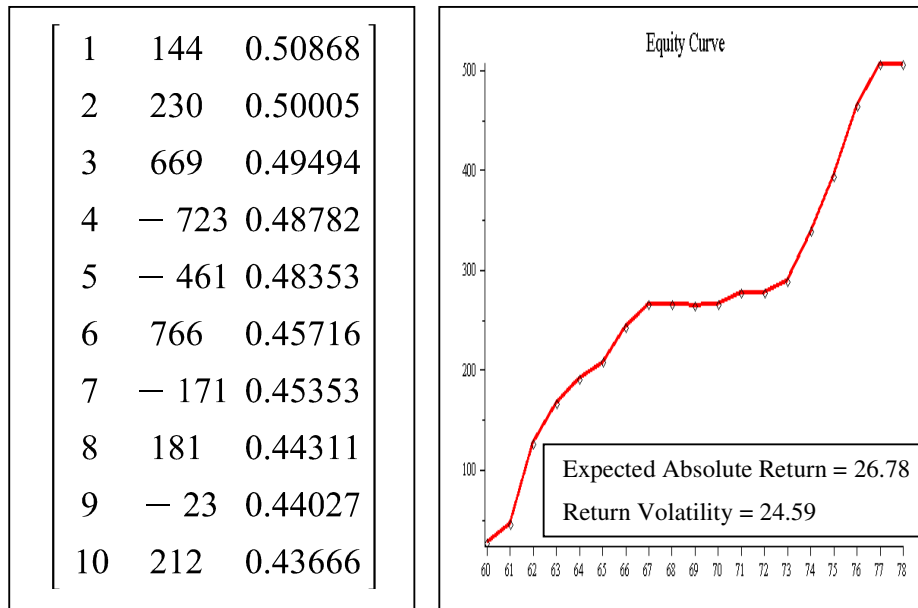


Exhibit 3. *Parameter Sensitivity and Profit / Loss*

We can now make our investment strategy a bit more sophisticated. We start by ranking our stocks according to the highest absolute risk adjusted return given a starting point and an end point. For this example we assume that the starting point is period 30 and the end point is period 60. Since we use highest absolute risk adjusted return both long and short positions are allowed. We then take a position in the 10 highest ranking stocks and calculate returns according to a trailing exit point. The exit point for this initial example is happening somewhere between observation 60 and observation 78, which is the last observation as seen in appendix-4. This whole optimization process is repeated for a 30 period rolling window i.e. period 31 to 61, 32 to 62, 33 to 63 etc. The result is the equity curve that can be seen in Exhibit-4. We can see that the equity curve is smooth and upward sloping. The annual percentage increase is 8.3 percent given the cost of the portfolio at each point in time and the absolute return. The drawdown with implementing such a strategy would be the transaction cost of having to rebalance the portfolio each month. However, due to the frequent rebalancing such strategy would be exposed to less systematic risk i.e. market risk.

*Exhibit 4. Dynamic Trading Strategy*

It is interesting to note that market neutral funds tend to have a more stabilizing effect on the financial markets as a whole, even though they can be quite vicious to prey on the already weak entities i.e. companies in a down trend. Trend following funds however tend to have a much more destabilizing impact i.e. when the market goes up everyone is on board and when the market goes down everyone is keen to short it. This leads to bubble behavior where prices deviate far from fundamental values. Such behavior creates opportunities if you know where to look. However, from a market neutral perspective such a behavior in combination highly interrelated financial markets can create problems because when the market goes down everything goes down with it. This leads to a switch-on-switch-off state where trend following funds tend to have an advantage and where market neutral funds will struggle to remaining market neutral and on the same hand being able to maximize expected return.

It is also interesting to note that changes in expected return can arise both due to firm specific risk and market risk. Further, one of the cornerstones of portfolio theory is that the more securities with negative cross correlated returns you have in a portfolio the lower portfolio risk you will have. The problem with this reasoning is that it only deals with portfolio returns. It ignores changes in expected return. When the market crashes most securities conditional an expected return becomes negative i.e. stocks are starting to trend downwards. Let's for example assume that you have a stop-loss investor. The piecewise Cumulative

Distribution Function (CDF) for such an investor given a zero mean, volatility of 20 and stop loss distance of $Stop$ is given by:

$$CDF = \begin{cases} 0 & x < Stop \\ \frac{1}{2} + \frac{erf\left(\frac{x * \sqrt{2}}{40}\right)}{2} & otherwise \end{cases}$$

Since the above CDF is not differentiable we need to convert it into a Heaviside function. We then take the derivative with respect to x in order to get the Probability Density Function (PDF). The PDF for the stop-loss investor is presented below:

$$PDF = \frac{\exp\left(-\frac{x^2}{800}\right) * \sqrt{2}}{40 * \sqrt{\pi}} + \frac{\delta * (x - Stop)}{2} + \frac{\delta * (x - Stop) * erf\left(\frac{x * \sqrt{2}}{40}\right)}{2} - \frac{\theta * (Stop - x) * \exp\left(-\frac{x^2}{800}\right) * \sqrt{2}}{40 * \sqrt{\pi}}$$

The expected return for the stop-loss investor, which is given by the $\text{integral}(\text{pdf} * x, x = -\infty, \infty)$ is given by:

$$ER = \frac{\left(1 + erf\left(\frac{Stop * \sqrt{2}}{40}\right)\right) * Stop}{2} + \frac{10 * \exp\left(-\frac{Stop^2}{800}\right) * \sqrt{2}}{\sqrt{\pi}}$$

The total return (TR) for the stop-loss investor is given by the below equation where ice is the probability of having a market crash i.e. iceberg risk, ER is the expected return for our stop loss distribution and n is the number of securities in the portfolio.

$$TR = (1 - ice) * (ER * n) + ice * (Stop * n)$$

If the market does not crash then our total return will be $ER * n$ which is equal to the expected return for each security multiplied by the number of securities in the portfolio. If the market crashes our return will be $Stop * n$ which is equal to the stop loss distance for all securities multiplied by the number of securities in the portfolio. I have in Exhibit-5 plotted the total return for different stops and probability of a market crash.

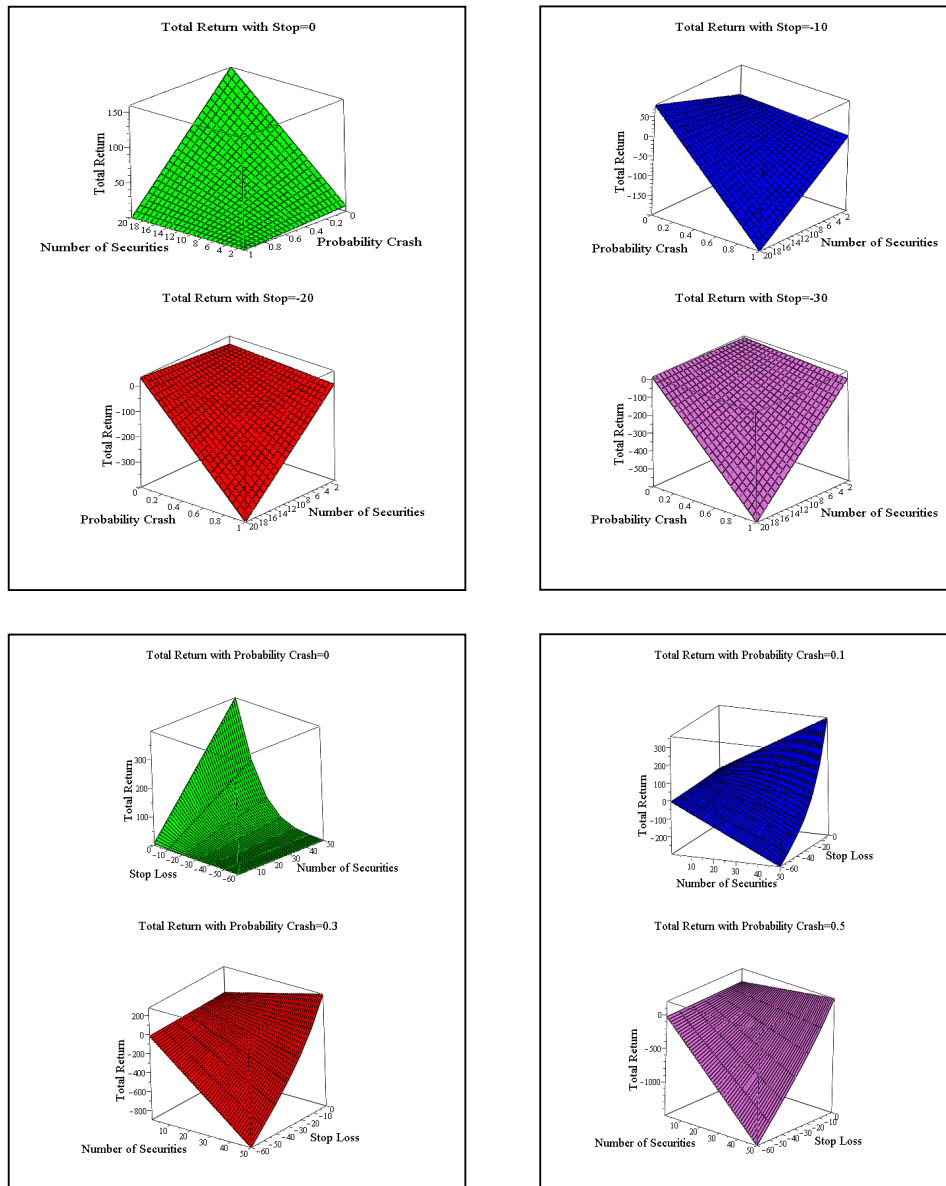


Exhibit 5. Total Return with Stop Loss and Probability Crash

We now assume that the probability of a market crash is represented by a random variable drawn from a Bernoulli distribution with probability p and where the total return is given by a piecewise function as seen below:

$$\begin{aligned} \text{Total Return} &= \begin{cases} ER * n & \text{ice} = 0 \\ Stop * n & \text{ice} = 1 \end{cases} \quad \text{where ice} \\ &= \text{RandomVariable}(\text{Bernoulli}(p)) \end{aligned}$$

We can now plot our total return and drawdown for a simulated portfolio as seen in Exhibit-6 below. We can see that our total return and portfolio drawdown is a linear and increasing function of the number of securities in the portfolio. These findings are important because it highlights the fact managing changes in conditional expected return is highly important even for a completely diversified portfolio. There are also exist many different interpretations of portfolio risk i.e. portfolio volatility vs. portfolio drawdown. Authors such as Seykota (2001) call portfolio drawdown portfolio “heat”. Portfolio heat is more specifically defined as the difference between the (price stock – minus the stop loss distance stock) *the number of stocks in portfolio*the position size. Portfolio heat answers the question: If all my stops were triggered at the same time what would my maximum loss be? Managing such portfolio heat becomes critical for any investor.

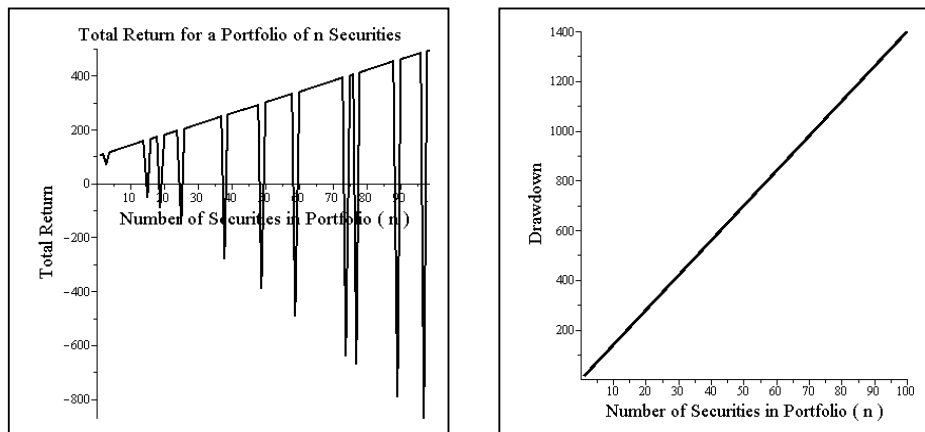


Exhibit 6. Total Return and Drawdown with Stop

Now imagine you have a diversified ten security long portfolio which has a positive expected return and low return variance. According to portfolio theory this would be a very attractive portfolio to hold because it has "low risk" and positive expected return. Now imagine you have a one security long portfolio which again has a positive expected return and low return variance i.e. trend following. Portfolio

theory will tell you that the first portfolio has lower risk because it is more diversified i.e. more return noise cancelation than the second. I do not agree that the first portfolio has lower risk. The first portfolio will have 10 times the risk of the second portfolio. This has to do with the fact that changes in expected return are 10 times more likely to take place in the first portfolio compared to the second portfolio. CAPM acknowledge this and suggests that we should balance long and short position in our portfolio so we can eliminate market risk i.e. changes in conditional expected return.

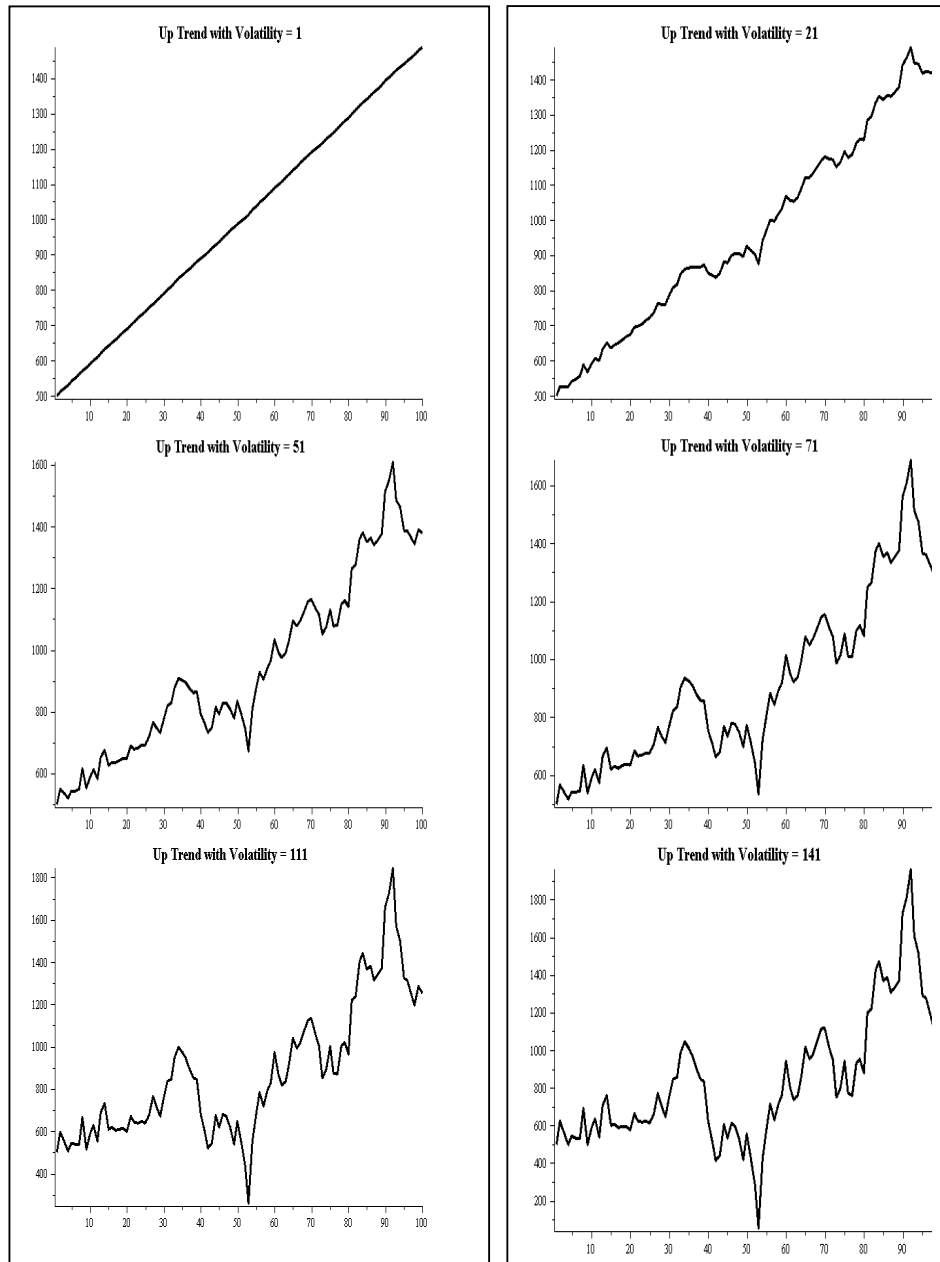
The problem is however that the returns are reduced as well i.e. when a long position is starting to lose money a short position is starting to make money hence a zero profit. This means that the CAPM does not really solve the problem. Now VaR has two components portfolio variance and conditional expected return. The way VaR works is that it assumes that the investor is educated enough to rebalance the portfolio frequently to eliminate risk due to changes in conditional expected return. The problem is that a return distribution or VaR will not tell you anything about how likely you are to experience changes in conditional expected return i.e. how often you need to rebalance the portfolio. This is a problem because if you had the option you would have preferred an ever increasing price trend over changes in conditional expected return any day. The frequency of changes in conditional expected return is very easy to estimate historically i.e. trending securities but not so easy to forecast. How do you forecast the next Bernard Madoff ? A person that have produced stable “returns” for a long time and suddenly blows up. Mandelbrot (1963) and Taleb (2007) go even further by arguing that not only can we not forecast changes in expected return but we cannot even guarantee that the returns will be normality distributed. Fat tail distributions are a very real empirical phenomenon. We started this paper by investigating a simple trading strategy where we entered a long position in all stocks that were available and we exited such positions by using a simple trailing stop loss. We say that such strategy had a positive expected return. We then discussed a simple trend following strategy where we took a long/short position in the stocks which had the highest risk adjusted returns. The strategy was then reiterated as a new observation became available to make sure that we were holding the stocks with most favorable risk adjusted returns. Diversification and cross correlation was ignored in order to simplify the analysis. All exits where done by using a trailing stop loss. This investment strategy was also highly successful with a smooth and increasing equity curve. The benefit with using a short rolling window instead of using a large rolling window in our risk adjusted return ranking is that the trading strategy easier picks up changes in expected return. If we use too much historical data the most recent observation, which are the most important once, get lost in the data. This results in that the signals are lost in a sea of noise. However, if we use a too small rolling window we might pick up a lot of noise instead of actual long term trends. It is important to find a good balance. Trend following strategies have received a lot of attention due to their promise of high and stable returns. This paper confirms that trend following investment strategies indeed represent a valid investment philosophy. It is important to note that trend following is highly integrated with

risk management i.e. trailing stop losses. A trend following strategy without a trailing stop loss has little value because it is difficult to determine when change in expected returns has taken place. One reason why trend following strategies are so successful might be the non predictive nature of trend following. A trend following trader simply react to market movements without trying to make any explicit assumptions about the future.

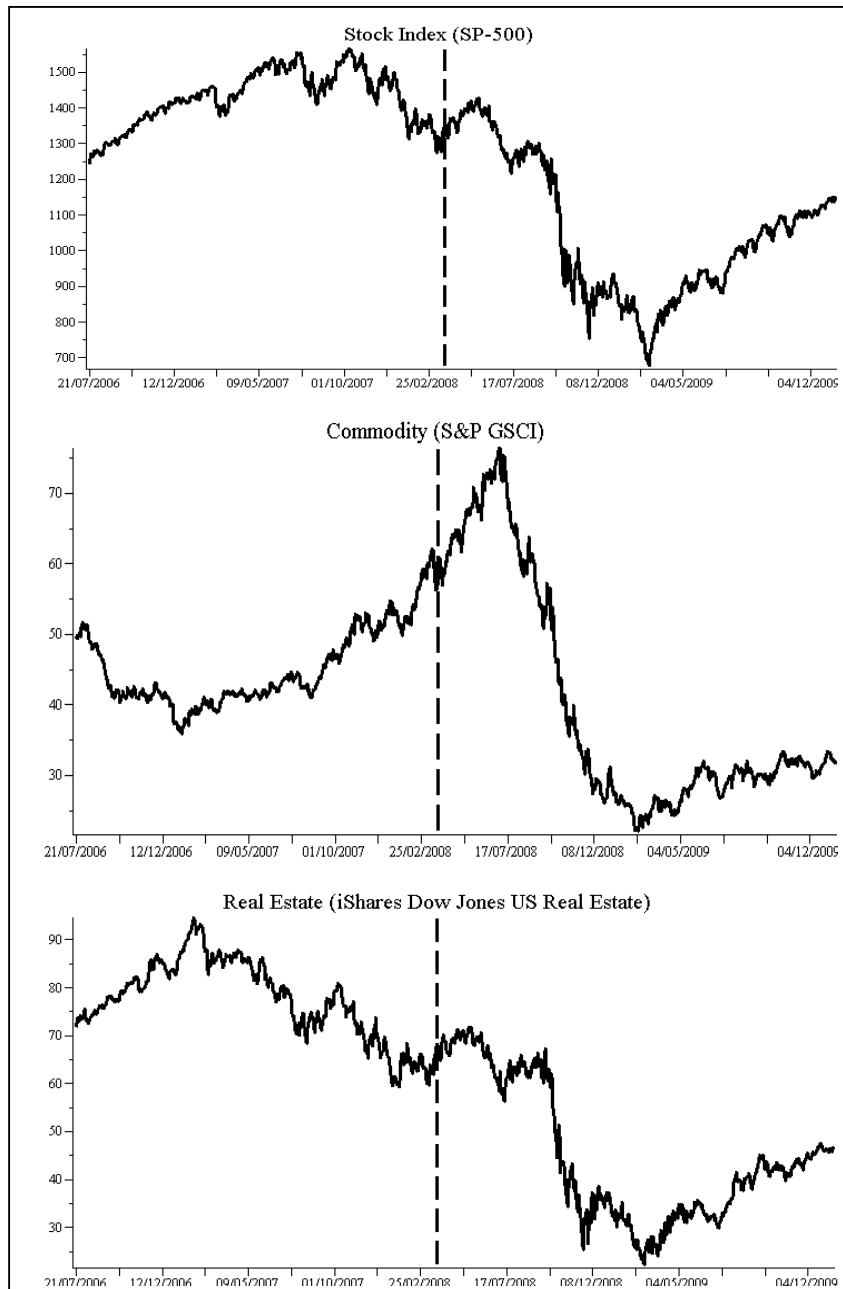
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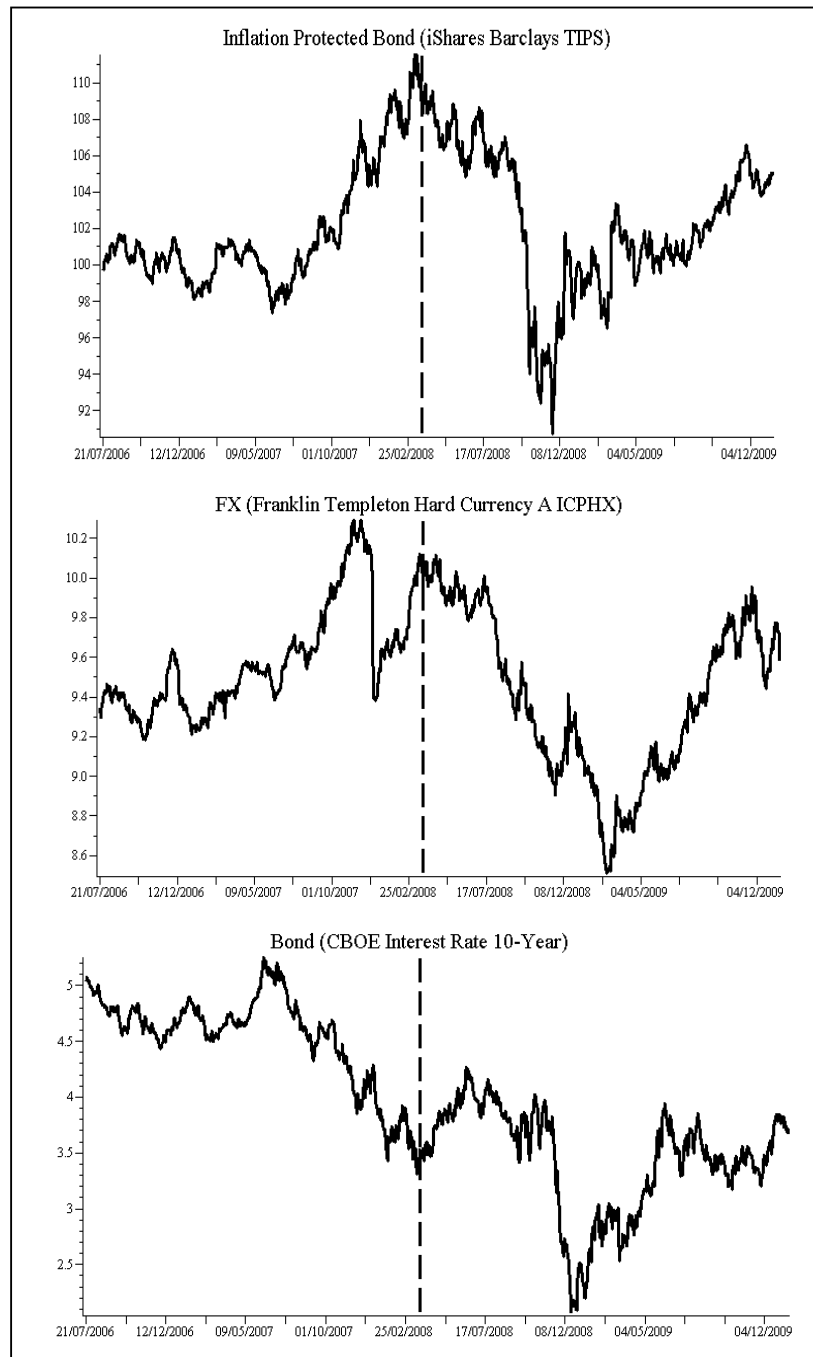
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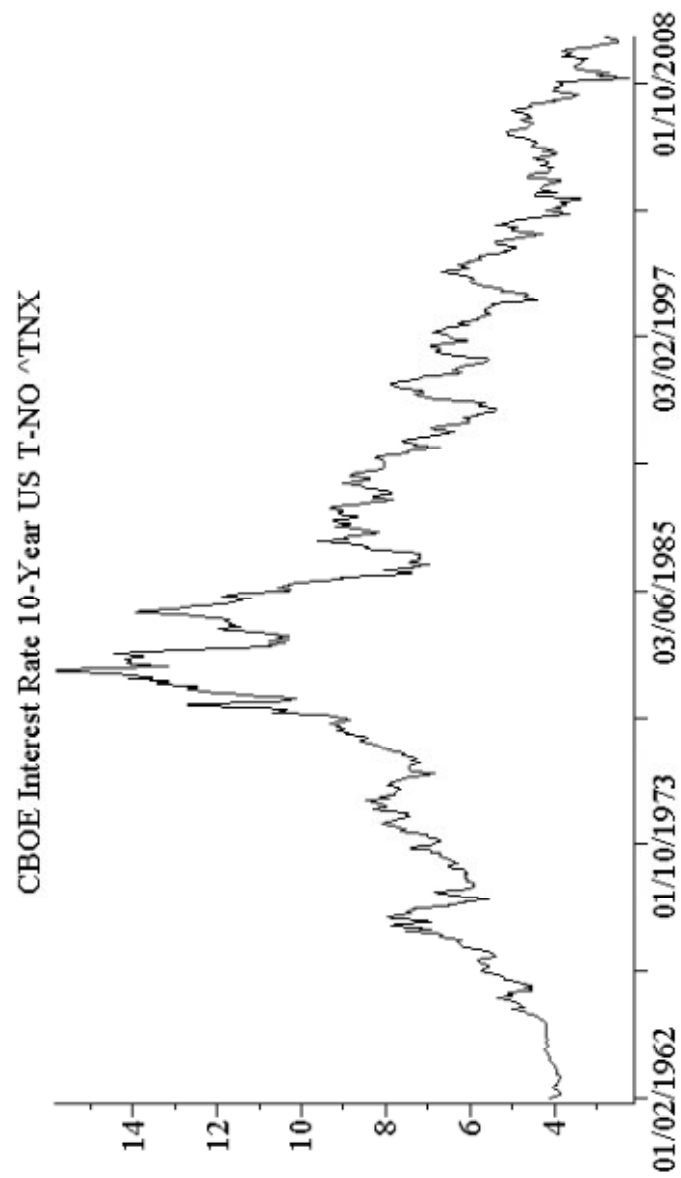
APPENDIX-1 TRENDS AND RETURN VOLATILITY



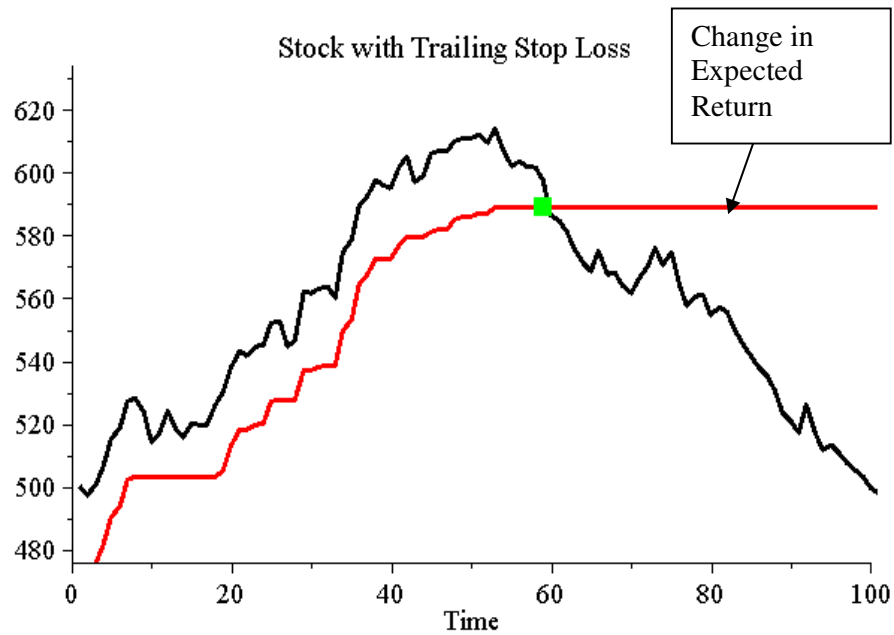
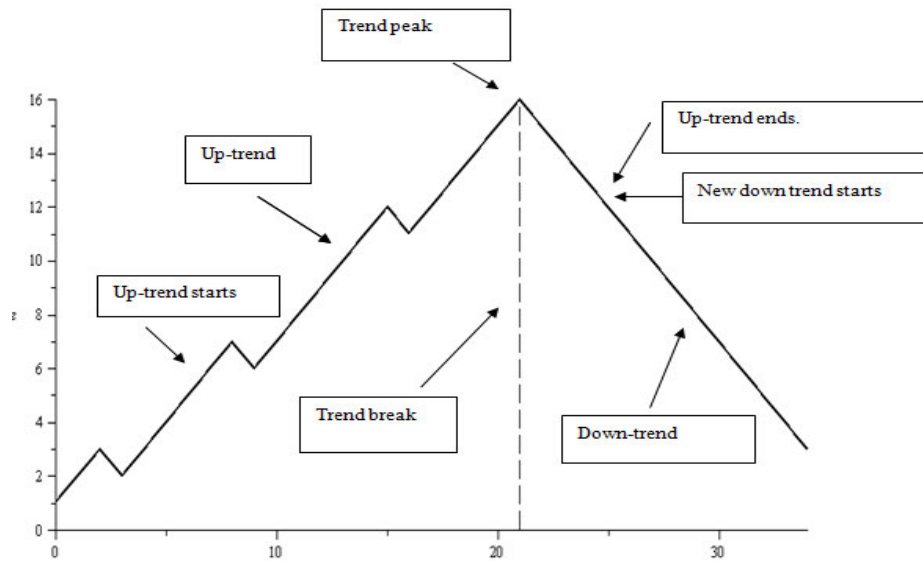
APPENDIX-2 THE 2008/2009 FINANCIAL CRISES BRANDED THE “CREDIT CRUNCH”.



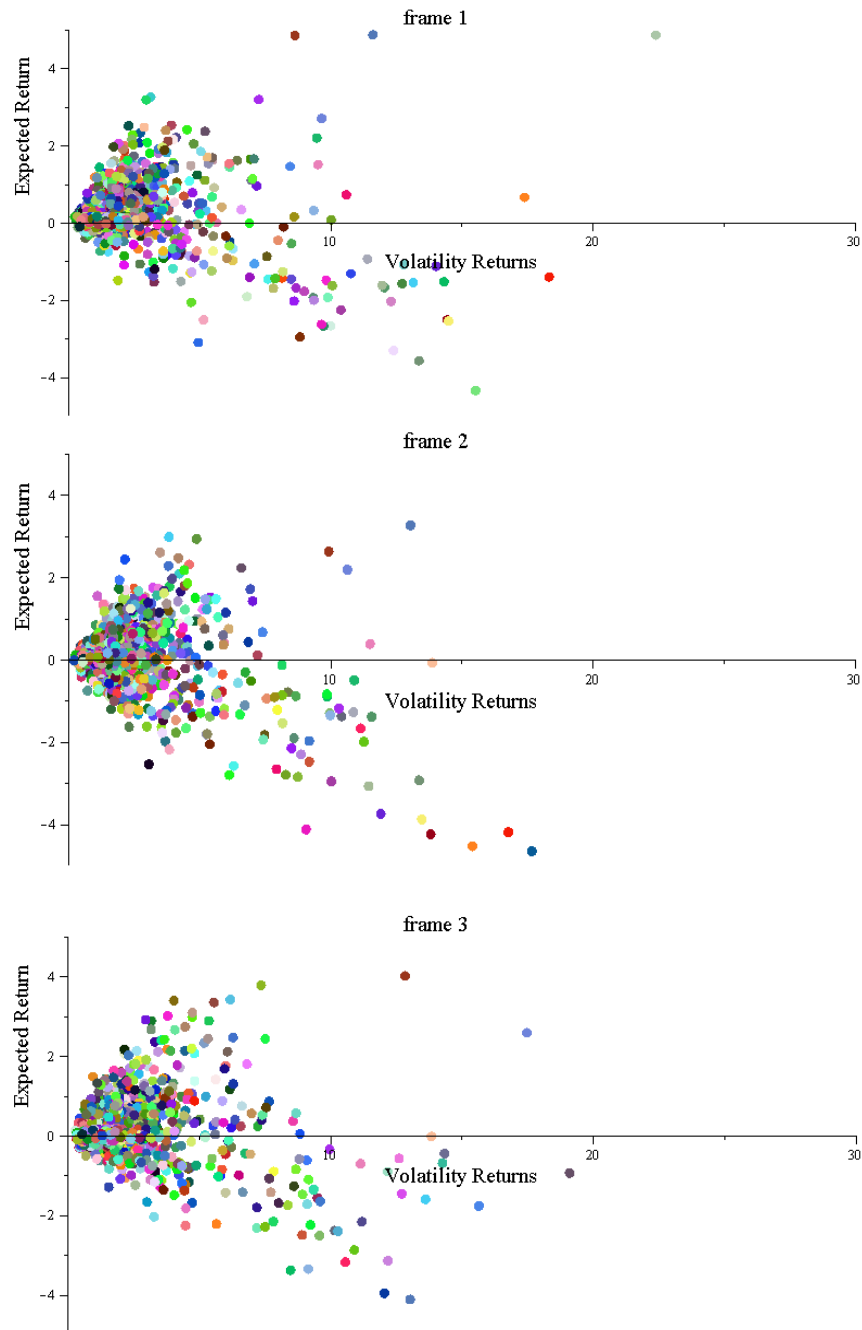




APPENDIX-3 THE DEFINITION OF A PRICE TREND



APPENDIX-4 RISK ADJUSTED RETURNS



APPENDIX-5 TRAILING STOP LOSS TRADING

